

Phụ lục: Kết quả Chi tiết từng Instance (Quy mô _21)

Bảng dưới đây trình bày giá trị Tốt nhất (Best) và Trung bình (Mean) của các hàm mục tiêu Z_1 (Số xe), Z_2 (Quãng đường), Z_3 (Workload Gini), Z_4 (Thời gian dài nhất) cùng thời gian chạy tính bằng giây cho từng thuật toán, thống kê trên 10 lần chạy (seeds).

Bảng 1: Chi tiết Kết quả Từng Instance (Quy mô _21)

Instance	Variant	Best Z_1	Mean Z_1	Best Z_2	Mean Z_2	Best Z_3	Mean Z_3	Best Z_4	Mean Z_4	Runtime
c101_21	FULL	12	12.00	1047.73	1051.75	0.0231	0.0284	1174.55	1175.34	285.3
c101_21	No-EQ	12	12.00	1043.45	1048.58	0.0707	0.0886	1174.55	1174.55	385.3
c102_21	FULL	11	11.00	1047.14	1054.99	0.0131	0.0174	1174.55	1180.53	284.3
c102_21	No-EQ	11	11.00	1037.12	1049.47	0.0264	0.0506	1174.55	1176.19	586.3
c103_21	FULL	10	10.90	991.16	1015.43	0.0136	0.0179	1174.55	1189.20	285.3
c103_21	No-EQ	10	10.40	990.52	998.32	0.0212	0.0381	1174.55	1185.95	564.3
c104_21	FULL	10	10.00	900.48	933.12	0.0049	0.0079	1166.58	1182.46	282.3
c104_21	No-EQ	10	10.00	895.60	907.11	0.0227	0.0326	1165.11	1168.36	457.3
c105_21	FULL	11	11.00	1055.83	1066.18	0.0106	0.0164	1132.27	1144.15	282.3
c105_21	No-EQ	11	11.00	1052.19	1058.51	0.0283	0.0324	1118.42	1130.19	614.3
c106_21	FULL	11	11.00	1041.46	1049.79	0.0165	0.0197	1109.14	1142.27	283.3
c106_21	No-EQ	11	11.00	1015.88	1039.44	0.0218	0.0386	1105.81	1123.72	706.3
c107_21	FULL	11	11.00	1032.73	1050.92	0.0091	0.0157	1132.75	1134.45	285.3
c107_21	No-EQ	11	11.00	1018.28	1040.98	0.0291	0.0384	1128.31	1133.22	774.3
c108_21	FULL	11	11.00	1015.93	1026.70	0.0110	0.0163	1116.84	1122.89	285.3
c108_21	No-EQ	10	10.90	1003.39	1019.66	0.0234	0.0396	1078.28	1118.04	808.3
c109_21	FULL	10	10.50	981.49	1010.52	0.0077	0.0203	1110.49	1157.67	285.3
c109_21	No-EQ	10	10.00	956.40	998.21	0.0210	0.0292	1161.72	1180.24	471.3
c201_21	FULL	4	4.00	629.95	629.95	0.0031	0.0059	3229.66	3229.66	228.3
c201_21	No-EQ	4	4.00	629.95	629.95	0.1083	0.1236	3229.66	3229.66	92.3
c202_21	FULL	4	4.00	629.95	630.34	0.0017	0.0035	3229.66	3229.66	254.3
c202_21	No-EQ	4	4.00	629.95	630.34	0.0824	0.1202	3229.66	3229.66	132.3
c203_21	FULL	4	4.00	629.95	631.42	0.0011	0.0025	3229.66	3229.66	275.3
c203_21	No-EQ	4	4.00	629.95	631.54	0.0611	0.1179	3229.66	3229.66	164.3
c204_21	FULL	4	4.00	629.95	635.22	0.0014	0.0023	3229.66	3229.66	285.3
c204_21	No-EQ	4	4.00	629.95	633.29	0.1098	0.1286	3229.66	3229.66	182.3
c205_21	FULL	4	4.00	629.95	629.95	0.0006	0.0035	3069.66	3078.68	239.3
c205_21	No-EQ	4	4.00	629.95	629.95	0.0492	0.0862	3069.66	3085.21	132.3
c206_21	FULL	4	4.00	629.95	629.95	0.0008	0.0030	2923.00	2962.26	237.3
c206_21	No-EQ	4	4.00	629.95	629.95	0.0037	0.0551	2923.00	2963.09	192.3
c207_21	FULL	4	4.00	629.95	629.95	0.0010	0.0027	2906.12	2947.46	244.3
c207_21	No-EQ	4	4.00	629.95	629.95	0.0080	0.0535	2906.12	2934.24	196.3
c208_21	FULL	4	4.00	629.95	629.95	0.0014	0.0036	2843.13	2871.51	257.3
c208_21	No-EQ	4	4.00	629.95	629.95	0.0053	0.0325	2839.83	2863.32	199.3
r101_21	FULL	18	18.80	1629.08	1646.91	0.0364	0.0485	224.69	225.97	281.3
r101_21	No-EQ	18	18.70	1620.68	1637.28	0.0695	0.0946	221.81	224.93	564.3
r102_21	FULL	16	16.70	1459.74	1496.57	0.0324	0.0436	220.91	223.90	285.3
r102_21	No-EQ	16	16.50	1441.85	1466.09	0.0622	0.0984	219.86	224.06	786.3
r103_21	FULL	14	14.00	1271.23	1288.94	0.0198	0.0283	222.48	222.92	285.3
r103_21	No-EQ	14	14.00	1248.90	1264.57	0.0609	0.0819	219.86	221.34	719.3
r104_21	FULL	12	12.00	1107.16	1128.31	0.0097	0.0186	219.86	223.04	285.3
r104_21	No-EQ	11	11.90	1069.94	1086.95	0.0481	0.0645	219.86	220.90	917.3
r105_21	FULL	15	15.30	1412.79	1445.47	0.0218	0.0347	216.84	221.74	285.3

Tiếp tục ở trang

Bảng 1 – Tiếp tục từ trang trước

Instance	Variant	Best Z_1	Mean Z_1	Best Z_2	Mean Z_2	Best Z_3	Mean Z_3	Best Z_4	Mean Z_4	Runtime
r105_21	No-EQ	15	15.00	1372.29	1394.80	0.0457	0.0798	218.85	222.39	649.5
r106_21	FULL	14	14.00	1302.99	1325.98	0.0147	0.0301	220.98	224.83	285.5
r106_21	No-EQ	14	14.00	1281.45	1309.59	0.0354	0.0623	217.25	220.83	782.5
r107_21	FULL	12	12.10	1149.35	1195.12	0.0136	0.0254	221.67	225.44	285.5
r107_21	No-EQ	12	12.10	1124.40	1150.01	0.0278	0.0446	217.91	222.60	933.5
r108_21	FULL	11	11.50	1054.07	1108.69	0.0060	0.0127	217.72	222.41	285.5
r108_21	No-EQ	11	11.10	1050.69	1066.14	0.0178	0.0368	218.42	222.32	704.5
r109_21	FULL	13	13.10	1254.19	1297.07	0.0137	0.0286	222.22	225.88	285.5
r109_21	No-EQ	13	13.00	1211.58	1233.74	0.0536	0.0728	222.76	224.53	896.5
r110_21	FULL	12	12.00	1120.35	1154.53	0.0112	0.0166	219.83	223.47	285.5
r110_21	No-EQ	12	12.00	1103.22	1119.40	0.0490	0.0645	217.89	220.70	839.5
r111_21	FULL	12	12.00	1135.22	1178.30	0.0156	0.0232	221.98	224.82	285.5
r111_21	No-EQ	12	12.00	1117.43	1132.84	0.0331	0.0512	221.96	222.91	986.5
r112_21	FULL	11	11.60	1054.61	1091.15	0.0161	0.0213	219.48	223.46	285.5
r112_21	No-EQ	11	11.20	1015.62	1053.12	0.0298	0.0413	214.51	220.84	735.5
r201_21	FULL	4	4.00	1182.66	1198.64	0.0012	0.0035	927.39	931.89	285.5
r201_21	No-EQ	4	4.00	1166.86	1181.80	0.0241	0.0803	922.36	931.39	330.5
r202_21	FULL	3	3.00	1061.98	1087.52	0.0001	0.0009	946.94	954.66	285.5
r202_21	No-EQ	3	3.00	1064.54	1077.06	0.0063	0.0403	943.73	950.09	444.5
r203_21	FULL	3	3.00	929.99	956.62	0.0002	0.0015	945.42	955.42	285.5
r203_21	No-EQ	3	3.00	903.33	919.89	0.0088	0.0265	932.91	945.58	506.5
r204_21	FULL	3	3.00	739.49	751.98	0.0004	0.0022	905.80	922.44	285.5
r204_21	No-EQ	3	3.00	720.10	733.46	0.0086	0.0467	901.41	907.62	630.5
r205_21	FULL	3	3.00	1001.99	1024.72	0.0001	0.0010	897.73	902.31	285.5
r205_21	No-EQ	3	3.00	1001.06	1012.43	0.0056	0.0208	891.10	901.45	376.5
r206_21	FULL	3	3.00	931.49	953.51	0.0002	0.0008	881.29	894.63	285.5
r206_21	No-EQ	3	3.00	928.40	947.16	0.0030	0.0328	884.04	891.81	425.5
r207_21	FULL	3	3.00	835.38	856.09	0.0003	0.0013	842.49	862.56	285.5
r207_21	No-EQ	3	3.00	816.56	831.37	0.0065	0.0386	843.02	847.70	682.5
r208_21	FULL	2	2.00	743.66	764.56	0.0000	0.0002	910.42	927.82	264.5
r208_21	No-EQ	2	2.00	737.53	750.45	0.0001	0.0016	906.43	913.82	433.5
r209_21	FULL	3	3.00	875.71	901.27	0.0001	0.0006	859.37	864.93	285.5
r209_21	No-EQ	3	3.00	877.86	893.22	0.0348	0.0557	859.39	860.53	464.5
r210_21	FULL	3	3.00	850.43	875.24	0.0001	0.0008	917.36	918.24	285.5
r210_21	No-EQ	3	3.00	851.55	864.41	0.0054	0.0365	917.36	918.36	419.5
r211_21	FULL	3	3.00	774.45	799.65	0.0007	0.0016	756.85	789.87	285.5
r211_21	No-EQ	3	3.00	780.83	790.95	0.0019	0.0385	761.15	775.68	570.5
rc101_21	FULL	16	16.90	1734.69	1776.78	0.0335	0.0447	230.10	233.38	285.5
rc101_21	No-EQ	16	16.40	1699.48	1737.83	0.0647	0.0850	230.10	233.27	629.5
rc102_21	FULL	15	15.30	1577.32	1616.82	0.0287	0.0429	230.71	233.18	285.5
rc102_21	No-EQ	15	15.00	1551.54	1584.93	0.0651	0.0924	232.87	233.23	546.5
rc103_21	FULL	13	13.30	1383.63	1435.17	0.0168	0.0318	231.51	235.45	285.5
rc103_21	No-EQ	13	13.00	1359.41	1389.16	0.0511	0.0740	230.14	233.99	644.5
rc104_21	FULL	11	11.80	1235.97	1273.22	0.0094	0.0137	230.88	234.68	276.5
rc104_21	No-EQ	11	11.80	1203.06	1235.44	0.0452	0.0639	227.49	233.42	538.5
rc105_21	FULL	14	14.90	1496.85	1545.61	0.0203	0.0343	224.86	228.05	285.5
rc105_21	No-EQ	14	14.50	1465.20	1503.08	0.0602	0.0854	223.08	225.96	581.5
rc106_21	FULL	14	14.00	1446.48	1475.98	0.0206	0.0283	224.16	229.73	285.5
rc106_21	No-EQ	14	14.00	1429.37	1450.81	0.0591	0.0770	226.10	227.00	730.5

Tiếp tục ở trang

Bảng 1 – Tiếp tục từ trang trước

Instance	Variant	Best Z_1	Mean Z_1	Best Z_2	Mean Z_2	Best Z_3	Mean Z_3	Best Z_4	Mean Z_4	Runtime
rc107_21	FULL	12	12.00	1278.14	1296.97	0.0147	0.0192	227.23	230.30	284.5
rc107_21	No-EQ	12	12.00	1267.26	1280.42	0.0356	0.0460	226.20	228.63	539.5
rc108_21	FULL	11	11.40	1183.12	1232.48	0.0123	0.0172	224.58	231.43	279.5
rc108_21	No-EQ	11	11.00	1189.79	1203.35	0.0173	0.0305	230.01	233.18	449.5
rc201_21	FULL	4	4.00	1461.82	1485.38	0.0006	0.0022	915.95	926.25	285.5
rc201_21	No-EQ	4	4.00	1463.95	1478.01	0.0091	0.0274	905.86	916.86	376.5
rc202_21	FULL	4	4.00	1267.37	1302.17	0.0008	0.0027	896.43	904.94	285.5
rc202_21	No-EQ	4	4.00	1243.33	1263.16	0.0109	0.0608	879.71	894.43	515.5
rc203_21	FULL	3	3.00	1092.09	1136.10	0.0001	0.0007	885.70	897.93	285.5
rc203_21	No-EQ	3	3.00	1087.18	1110.39	0.0102	0.0317	885.70	893.73	364.5
rc204_21	FULL	3	3.00	901.94	919.01	0.0002	0.0007	839.48	850.44	280.5
rc204_21	No-EQ	3	3.00	892.30	899.66	0.0173	0.0538	839.48	845.68	248.5
rc205_21	FULL	4	4.00	1191.33	1218.23	0.0004	0.0023	888.17	888.17	285.5
rc205_21	No-EQ	4	4.00	1177.19	1205.06	0.0365	0.0810	888.17	888.17	338.5
rc206_21	FULL	3	3.80	1110.55	1168.81	0.0006	0.0030	825.36	843.34	284.5
rc206_21	No-EQ	3	3.60	1099.03	1175.65	0.0145	0.0440	819.42	848.44	388.5
rc207_21	FULL	3	3.00	1008.55	1048.89	0.0001	0.0008	839.29	851.56	283.5
rc207_21	No-EQ	3	3.00	996.62	1029.52	0.0064	0.0249	828.76	853.75	366.5
rc208_21	FULL	3	3.00	844.67	882.35	0.0005	0.0011	754.45	775.46	285.5
rc208_21	No-EQ	3	3.00	837.66	868.75	0.0123	0.0295	744.74	763.48	397.5